

The transshipment problem

Optimality conditions

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Optimization: principles and algorithms

Optimality conditions

The transshipment problem is a linear optimization problem, where the constraints have a special structure.

We analyze in this video the dual problem and the dual variables.

We show the optimality conditions called “complementarity slackness”.

Transshipment problem in standard form

$$\min_{x \in \mathbb{R}^n} \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}$$

subject to

$$\sum_{j|(i,j) \in \mathcal{A}} x_{ij} - \sum_{k|(k,i) \in \mathcal{A}} x_{ki} = s_i, \forall i \in \mathcal{N},$$
$$x_{ij} \geq 0, \forall (i,j) \in \mathcal{A}.$$

subject to

$$\min_{x \in \mathbb{R}^n} c^T x$$

$$Ax = s,$$
$$x \geq 0.$$

Lagrangian

$$\min_{x \in \mathbb{R}^n} \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}$$

subject to

$$\sum_{j|(i,j) \in \mathcal{A}} x_{ij} - \sum_{k|(k,i) \in \mathcal{A}} x_{ki} = s_i, \forall i \in \mathcal{N},$$
$$x_{ij} \geq 0, \forall (i,j) \in \mathcal{A}.$$

Lagrangian: $L(x, \lambda, \mu) =$

$$\sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}$$
$$+ \sum_{i \in \mathcal{N}} \lambda_i \left(\sum_{j|(i,j) \in \mathcal{A}} x_{ij} - \sum_{k|(k,i) \in \mathcal{A}} x_{ki} - s_i \right)$$
$$- \sum_{(i,j) \in \mathcal{A}} \mu_{ij} x_{ij}.$$

First order optimality conditions (KKT)

$$\begin{aligned} L(x, \lambda, \mu) = & \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij} \\ & + \sum_{i \in \mathcal{N}} \lambda_i \left(\sum_{j | (i,j) \in \mathcal{A}} x_{ij} - \sum_{k | (k,i) \in \mathcal{A}} x_{ki} - s_i \right) \\ & - \sum_{(i,j) \in \mathcal{A}} \mu_{ij} x_{ij}, \end{aligned}$$
$$\begin{aligned} \frac{\partial L}{\partial x_{ij}} = c_{ij} + \lambda_i - \lambda_j - \mu_{ij} &= 0. \\ \mu_{ij} &= c_{ij} + \lambda_i - \lambda_j \\ \mu_{ij} \geq 0 \text{ and } \mu_{ij} x_{ij} &= 0 \end{aligned}$$

Comments

Mention that only the differences of λ matters.

Complementarity slackness

For all (i, j)

$$c_{ij} + \lambda_i - \lambda_j \geq 0.$$

For all (i, j) such that $x_{ij} > 0$

$$c_{ij} + \lambda_i - \lambda_j = 0.$$

Sufficient and necessary optimality conditions.

Summary

Simple optimality conditions

A dual variable is associated with each node in the network.

Only the difference between to adjacent nodes matters

The dual variables are often called “labels”.